



BIT 5424 – Data Collection Worksheet

GROUP 04

Team members' names:

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1. Big Idea (write a sentence or two below describing the overall reason for your project):

The overall goal of this project is to understand how safe the Virginia Tech campus actually feels for students and to identify the biggest gaps. By identifying patterns in student experiences, the project aims to give campus leaders clear evidence they can use to improve conditions and make decisions that better support students' daily needs. The final analysis also compares student perceptions with VTPD records to distinguish areas where student concerns are supported by official data from those where perception and reality diverge, especially regarding nighttime safety and confidence in reporting.



2. Your data collection table (fill out this table, use the template on Canvas to help you). Think of each row in the table as a dashboard in your Tableau data story. You can add rows to the table.

Inquiry Question (Curiosity) ¹	Information Needed to Answer Inquiry Question	Measurement Indicator	Data Sources	Data Collection Methods	Data Collectors ²	Timing and Location of Data Collection
How do perceptions of safety on and around the Virginia Tech campus differ across student demographic groups, such as gender, academic role, and disability status, and which groups report feeling most at risk by time of day?	Student demographics (gender, academic role, disability status); perceived safety ratings by time of day (day vs. night); residence location (on/off campus)	% distribution of respondents by demographic group; average safety rating by time of day per demographic group; % of each group reporting feeling unsafe at night	Team 4's Survey	Conduct a survey of Virginia Tech students across multiple class standings, genders, and ethnicities	Team 4 (Yash, Nik, Tessa, Ritika, Mouni, Josh)	Spring 2026, Virginia Tech Campus, Internet Survey

¹ See the handout on Canvas for an example of what should go in each field.

² Data collectors = where you can get the data from (e.g., which websites) and who is going to be responsible for collecting that portion of the data.

Reference [1]: **VTPD Crime Logs** -
<https://police.vt.edu/crime-stats/crime-logs.html>

Reference [2]: **VT ASR** -
https://police.vt.edu/content/dam/police_vt_edu/clery-reports/2024%20ASR.pdf



Inquiry Question (Curiosity) ¹	Information Needed to Answer Inquiry Question	Measurement Indicator	Data Sources	Data Collection Methods	Data Collectors ²	Timing and Location of Data Collection
Are students aware of Virginia Tech's key safety resources and reporting mechanisms, and does greater familiarity with those resources correlate with higher confidence in how incidents are resolved?	Student awareness of VT safety resources; familiarity with reporting mechanisms; self-reported confidence in how incidents are resolved	% of students aware of each safety resource; % who have used each resource; average confidence in reporting score (1-5 scale); comparison of confidence ratings between students aware vs. unaware of resources	Team 4's Survey	Conduct a survey of Virginia Tech students across multiple class standings and genders, asking about resource awareness, usage, and confidence in incident resolution	Team 4 (Yash, Nik, Tessa, Ritika, Mouni, Josh)	Spring 2026, Virginia Tech Campus, Internet Survey
How do students' primary modes of transportation shape their experience of campus safety, and which commute methods are associated with the greatest perceived risk?	Primary commute mode (walking, bus, car, bike); perceived safety by commute mode; satisfaction with the safety of each transportation option	% of students using each commute mode; average safety rating by commute mode; % reporting safety concerns specific to each mode; comparison of safety ratings across walking, bus, car, and bike users	Team 4's Survey	Conduct a survey of Virginia Tech students across multiple class standings, genders, and ethnicities;	Team 4 (Yash, Nik, Tessa, Ritika, Mouni, Josh) VTPD crime logs accessed from reference [1]	Spring 2026, Virginia Tech Campus, Internet Survey VTPD: <i>Spring 2026; VTPD Monthly Crime Logs accessed online from reference [1]</i>

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Inquiry Question (Curiosity) ¹	Information Needed to Answer Inquiry Question	Measurement Indicator	Data Sources	Data Collection Methods	Data Collectors ²	Timing and Location of Data Collection
				Extract and tabulate incident records from VTPD Monthly Crime Logs by location and time of day		
Where on and around the Virginia Tech campus are crimes most geographically concentrated, and do those hotspots align with the locations students report feeling least safe?	Incident locations from VTPD crime log; student-reported locations where they feel least safe	Frequency and % share of incidents by location category; top incident locations by count; overlap between student-reported unsafe locations and VTPD-documented incident locations	VTPD Monthly Crime Logs	Geocode incident locations from VTPD Monthly Crime Logs; cross-reference with student survey responses on perceived unsafe locations	Team 4 (Yash, Nik, Tessa, Ritika, Mouni, Josh); VTPD crime logs accessed from reference [1]	Spring 2026, Virginia Tech Campus, Internet Survey; VTPD Monthly Crime Logs accessed online from reference [1]
When are crimes most likely to occur in terms of time of day and period of the year, and do those temporal patterns align with student perceptions of when the campus feels least safe?	Dates and times of occurrence from VTPD crime log; time of day categories (morning, noon, evening, night); monthly and seasonal trends across 2022-2026;	% of incidents by time of day category; % of incidents by month and season; year-over-year incident frequency trends;	VTPD Monthly Crime Logs; Virginia Tech 2024 Annual Security Report	Derive time of day categories from the VTPD occurrence time field; tabulate incidents by month and year;	Team 4 (Yash, Nik, Tessa, Ritika, Mouni, Josh) VTPD crime logs accessed from reference [1]	Spring 2026, Virginia Tech Campus, Internet Survey VTPD: <i>Spring 2026; VTPD Monthly Crime Logs accessed online from reference [1]</i>

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	student-reported perceptions of when campus feels least safe	comparison between VTPD temporal patterns and student perceptions of when campus feels least safe		compare temporal patterns with student survey responses on perceived unsafe times	VT 2024 Annual Security Report accessed via reference [2]	ASR: <i>Spring 2026; Virginia Tech 2024 Annual Security Report accessed online via reference [2]</i>
How are reported crimes on the Virginia Tech campus ultimately resolved, and do resolution patterns vary meaningfully across different offense types?	Disposition/resolution type for each incident (e.g., Cleared by Arrest, Referred to Student Conduct, Inactive); offense type associated with each case outcome	% of incidents by resolution type; breakdown of resolution types per offense category; comparison of resolution rates across the top offense types	VTPD Monthly Crime Logs	Extract and standardize disposition fields from VTPD Monthly Crime Logs; cross-reference with offense categories to identify resolution patterns	Team 4 (Yash, Nik, Tessa, Ritika, Mouni, Josh); VTPD crime logs accessed from reference [1]	Spring 2026; VTPD Monthly Crime Logs accessed online from reference [1]

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Inquiry Question (Curiosity) ¹	Information Needed to Answer Inquiry Question	Measurement Indicator	Data Sources	Data Collection Methods	Data Collectors ²	Timing and Location of Data Collection
Where do student perceptions of safety and confidence in reporting align with or diverge from ground reality as captured in the VTPD crime records, and what does that gap reveal for stakeholders?	Student-reported safety ratings by time of day (day vs. night); student confidence in issue resolution by academic role; VTPD incident counts by time of day; VTPD case resolution rates by offense type; top crime location from VTPD records	% of students feeling safe during the day and night; average confidence score by academic role; VTPD incident count by time of day; case resolution rate for the top 15 offenses; most frequently reported crime location; VTPD crimes at night and midnight as a raw count	Team 4's Survey; VTPD Monthly Crime Logs	Cross-reference survey safety ratings and confidence scores against VTPD incident counts by time of day and case resolution rates; identify top crime location from VTPD records	Team 4 (Yash, Nik, Tessa, Ritika, Mouni, Josh); VTPD crime logs accessed from reference [1]	Spring 2026, Virginia Tech Campus, Internet Survey; VTPD Monthly Crime Logs accessed online from reference [1]

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3. Describe how you will analyze (make meaning from) the collected data (write your plan below):

Think about what relationships you hope to uncover in the data and what different audiences you expect to present your data to.

We will analyze the data by looking for patterns among students' demographics, their campus use, and the types of safety issues they reported. We want to see which problems arise most often, where they occur, and whether certain groups experience them more than others. Our approach proceeds in three layers: first, descriptive statistics and frequency distributions across all survey items; second, subgroup segmentation by gender, disability status, and commute mode to identify differential experiences; and third, cross-referencing survey patterns with VTPD crime log data to assess alignment between perceived and documented risk. Because some subgroup sizes are small, especially for non-binary respondents, subgroup comparisons will be treated as directional rather than statistically definitive.

For different audiences, we will highlight the results that matter most to them. For example, transportation groups will see trends about buses and transit areas, while campus planners and safety stakeholders will see issues related to lighting, location-based risk, reporting confidence, and high-concern routes.

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